

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

Degree, Semester& Branch: VII Semester B.E. EEE

Course Code & Title: EE6008 Microcontroller Based System Design

Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic: Introduction to PIC Microcontroller

Name of the Innovative Practice: Animated Videos

Date & Time: 06.07.2019 &15 Minutes

ICT Tool Used: LCD projector, Personal Computer, Speakers

Description:

Over the past few years, videos are being widely used in classrooms for supporting a teacher's curriculum and helping students learn the material faster than ever. Research shows that 94% of the teachers have effectively used videos during the academic year and they have found video learning quite effective, it is even better than teaching students through traditional text-books.

Majority of part of the human brain is devoted towards processing the visual information. Brain responds to visuals fast, better than text or any other kind of learning material. Remembering stuff from the picture is retained in the mind for a longer time. Through videos, students get to process information fast.

Goals (Learning Outcomes):

- The students can gain knowledge in PIC Microcontroller.

Use of appropriate methods:

Justification for choosing Animated Videos:

PIC Microcontroller used in many applications in industries, automobiles and home appliances etc., Animated video shows, how PIC Microcontroller used in various applications, it gives exposure about embedded based products.

Effective presentation

Details of the Implementation

The topic was first handled and many of the students were not able to imagine and visualize the operation and application of PIC Microcontroller. Then the video was shown which made the students to understand well about PIC Microcontroller and applications.

Innovative Teaching Method Execution

Embedded ARM Applications



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated by asking architecture of PIC Microcontroller in Internal Assessment Test – 1 most of the students attended the question and answered correctly.

Reflective Critique:

Benefits:

The students were more interested in learning the concepts if we visually show the videos. They will be able to easily understand the application of PIC Microcontroller and motivates to do embedded system based projects.

Challenges:

It takes some time to choose relevant videos in PIC Microcontroller.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	3	2	2	-	-	2	-	2	-	2

CO1: The Students will be able to explain the functions of the blocks in PIC microcontroller.

References:

1. Furber,S., “ARM System on Chip Architecture” Addison Wesley trade Computer Publication, 2000.
2. <http://www.technologystudent.com/pics/picgen1.html>
3. <https://www.youtube.com/user/MicrochipTechnology>

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Degree, Semester& Branch: VII Semester B.E. EEE

Course Code & Title: EE6008 Microcontroller Based System Design

Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic: Introduction to PIC microcontroller Interrupts- External Interrupts

Name of the Innovative Practice: Crossword Puzzles

Date & Time: 22.07.2019& 20 Minutes

ICT Tool Used: LCD projector, Personal Computer, Speakers

Description:

Crossword puzzle technique is increase the interest of students in exam point of view. Crossword puzzle is advantageous in self-correcting method due to the length of each word and the overlap of each answer with other answers. The crossword puzzles activity module allows the teacher to design and build quizzes consisting of a large variety of Question types.

Goals (Learning Outcomes):

Learning Outcomes:

- The students can gain knowledge about interrupt in PIC Microcontroller.

Use of appropriate methods:

Justification for choosing quiz:

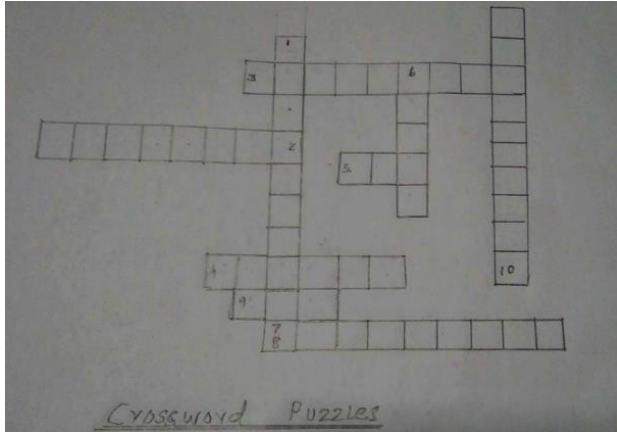
Hence this crossword puzzles activity helps the students to think in multiple dimensions and provide the answers.

Effective presentation

Details of the Implementation

The questions and crossword puzzle test format were displayed in the classroom and the students were asked to answer the questions from the knowledge of what they learnt in the classroom. At the end of the activity the correct answers were discussed.

Innovative Teaching Method Execution
Introduction to PIC microcontroller Interrupts- External Interrupts



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- Most of the questions were correctly answered by the students. The wrongly answered questions were discussed at the end of the class.

Reflective Critique:

Benefits:

- The students got a clear idea about this topic and they felt it was useful, they are eagerly participated.

Challenges:

- I planned this activity for 15 minutes and it took around 20 minutes to finish this activity.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	1	2	-	-	-	-	1	-	2	-	2

CO2: The Students will be able to apply the different Interrupts and Timers in PIC microcontroller.

References:

1. Peatman, J.B., "Design with PIC Micro Controllers" Pearson Education, 3rd Edition, 2004.
2. <https://blog.crosswordhobbyist.com/teaching-with-educational-crossword-puzzles-based-on-research>

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Innovative Practices Description

Degree, Semester& Branch: VII Semester B.E. EEE

Course Code & Title: EE6008 Microcontroller Based System Design

Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic: ARM programmer's model

Name of the Innovative Practice:Hardware Demonstration

Date & Time:03.09.2019& 15 Minutes

ICT Tool Used: LCD projector, Personal Computer, Speakers

Description:

A method demonstration is a teaching method used to communicate an idea with the aid of visuals such as flip charts, posters, power point, etc. In demonstration method, the teaching-learning process is carried in a systematic way. Demonstration often occurs when students are unable to understand theories concept.

Demonstrations help to raise student interest and reinforce memory retention because they provide connections between facts and real-world applications of those facts. Lectures, on the other hand, are often geared more towards factual presentation than connective learning.

Goals (Learning Outcomes):

- The students will able to understand the LPC 2148 ARM 7 Processor and its programming.

Use of appropriate methods:

Justification for choosing activity:

Theoretically students know the architecture and various pins of ARM processor, the hardware demonstration gives more exposure and easily understand various modules and functionality of ARM processor. Students come know how to embed program into ARM processor.

Effective presentation

Implementation (Plan & Execution) with Proof:

Innovative Teaching Method Execution
ARM programmer's model



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated by asking ARM Programmers model in Internal Assessment Test III – Around 80% of students answered correctly.

Reflective Critique:

Benefits:

- From this demonstration students easily understand the hardware architecture of ARM7TDMI, features and its applications.

Challenges:

- I planned to conduct this activity within 5 minutes. But it took around 15 minutes to finish this activity

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	1	2	1	2	-	-	1	-	2	-	2

CO4: The Students will be able to explain the programmer's model of ARM processor.

Reference:

1. Furber, S., "ARM System on Chip Architecture" Addison Wesley trade Computer Publication, 2000.
2. <http://infocenter.arm.com/help/index.jsp?topic=/com.arm.doc.ddi0234b/BABHJFCG.html>
3. http://ww1.microchip.com/downloads/en/DeviceDoc/DDI0029G_7TDMI_R3_trm

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Course Code & Title: EE6008 Microcontroller Based System Design

Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic:Architectural Support for Operating systems

Name of the Innovative Practice: Minute Paper

Date & Time: 19.09.2019& 5 Minutes

ICT Tool Used: LCD projector, Personal Computer, Speakers

Description:

The Minute Paper is a very commonly used classroom assessment technique. It really does take about a minute and, while usually used at the end of class, it can be used at the end of any topic discussion. Its major advantage is that it provides rapid feedback on whether the professor's main idea and what the students perceived as the main idea are the same. Additionally, Students must first organize their thinking to rank the major points and then decide upon a significant question. It is thus a very adaptable tool.

Goals (Learning Outcomes):

1. The students will be able to understand the MMU Register and Protection Registers in ARM Processor.
2. The students will be able to understand the concept of context switching.

Use of appropriate methods:

Justification for choosing the topic using One Minute Paper activity:

Architectural Support for Operating systems is important topic, it contains many subtopics. Students find difficult to remember all the registers in MMU and Production registers in Co-Processor 15 and its operations. Minute paper activity helps students to understand and differentiate various registers in CP15.

Effective presentation

Implementation (Plan & Execution) with Proof:

Minute papers are an effective way of involving all students in class simultaneously. It ensures equal participation of each and every class member, including anyone who may be too shy or fearful to participate orally. Each and every student is expected to participate and has something important to contribute no matter what their cultural background or prior level of academic preparedness.

Innovative Teaching Method Execution

Architectural Support for Operating systems - Minute paper

ARCHITECTURAL SUPPORT FOR OS M. KIRUBAH

OS provide an environment, several program run concurrently.

ARM coprocessors

on chip processor, using logical coprocessor number 15, control on chip cache, MMU, etc.

Reg 0

Implemented architecture fast number register

Reg 1

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
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Reg 2

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
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Memory Granularity

Section 1 MBs block of memory

953616105043

Memory Granularity:

Section: 1 MB block of memory

Large pages: 64 Kbyte blocks of memory. to individual 16 Kbyte subpages.

Small pages: 4 Kbyte blocks of memory to individual 1 Kbyte subpages.

Tiny pages: 1 Kbyte tiny pages

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The students felt easy to remember and recollect function of the MMU register and Protection register in Co-Processor 15.

Reflective Critique:

Benefits:

- Minute papers can improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

Challenges:

- As it is mentioned above that it takes one minute to conduct this activity. But, the students are failed to remember all equations in a given duration. They took five minutes to complete this activity.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	1	2	2	-	-	-	1	-	2	-	1

CO4: The Students will be able to explain the programmer's model of ARM processor.

Reference:

- Furber, S., "ARM System on Chip Architecture" Addison Wesley trade Computer Publication, 2000.
- <https://provost.tufts.edu/celt/files/MinutePaper.pdf>

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Course Code & Title: EE6008 Microcontroller Based System Design

Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic: Embedded ARM Applications

Name of the Innovative Practice: Animated Videos

Date & Time: 04.10.2019&15 Minutes

ICT Tool Used: LCD projector, Personal Computer, Speakers

Description:

Over the past few years, videos are being widely used in classrooms for supporting a teacher's curriculum and helping students learn the material faster than ever. Research shows that 94% of the teachers have effectively used videos during the academic year and they have found video learning quite effective, it is even better than teaching students through traditional text-books.

Majority of part of the human brain is devoted towards processing the visual information. Brain responds to visuals fast, better than text or any other kind of learning material. Remembering stuff from the picture is retained in the mind for a longer time. Through videos, students get to process information fast.

Goals (Learning Outcomes):

- The students can gain knowledge in various applications of ARM processors.

Use of appropriate methods:

Justification for choosing Animated Videos:

ARM processor used in many application in industries, Automobiles, Mobile phones, Cellular communication etc., Animated video shows, how ARM processor used in various applications, it gives exposure about embedded based products.

Effective presentation

Details of the Implementation

The topic was first handled and many of the students were not able to imagine and visualize the operation of ARM processor. Then the video was shown which made the students to understand well about ARM processor and applications of ARM processor in Mobile phone, Industry.

Innovative Teaching Method Execution	
Embedded ARM Applications	
	

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated by asking application of ARM Processor on Internal Assessment test III – Around 85% of students answered correctly.

Reflective Critique:

Benefits:

The students were more interested in learning the concepts if we visually show the videos. They will be able to easily understand the application of ARM processor and motivates to do embedded system based projects.

Challenges:

It takes some time to choose relevant videos in ARM embedded application.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	3	3	2	2	-	-	2	-	2	-	2

CO5: The Students will be able to analyze the applications of ARM processor.

References:

1. Furber,S., “ARM System on Chip Architecture” Addison Wesley trade Computer Publication, 2000
2. NCTEL- NITTTR Chandigarh Technology enabled Learning-
https://www.youtube.com/watch?v=dZy8Z6_BcWg

3. <https://www.microchip.com/design-centers/smartphone-accessory>, https://www.youtube.com/watch?v=avO6AQ_fF6Y